Name: Casey Tom | Student ID: 1230794853

Project 2 Part 1: K-Means Strategy 1

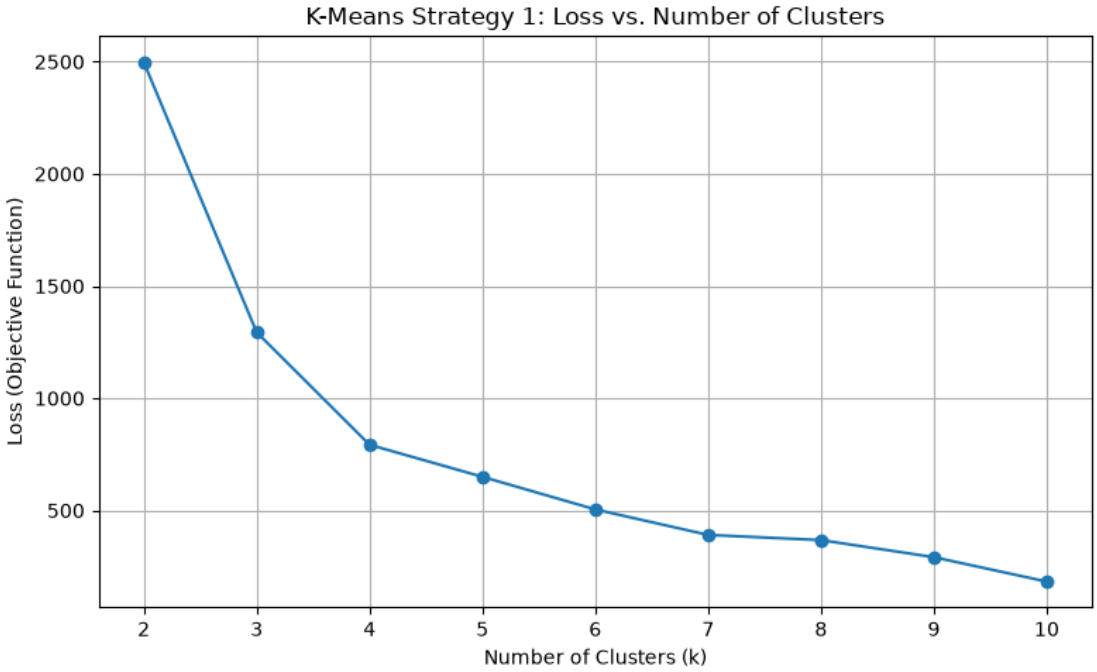
Project Analysis and Objective

The objective of Part 1 was to implement the K-means clustering algorithm using Strategy 1 and apply it to the provided dataset of 300 two-dimensional samples. The algorithm was evaluated for k = 2 through k = 10. For each value of k, Strategy 1 randomly selected the initial centroids from the dataset using the provided initial_S1() function, which used the last four digits of the student ID to determine the selected samples. This provided the prescribed Strategy 1 initialization while making the results reproducible for the assigned ID. For each k, every sample was assigned to the nearest centroid according to squared Euclidean distance. Each centroid was then recalculated as the mean of the samples assigned to its cluster. The assignment and centroid-update steps continued until the centroid coordinates were exactly unchanged between consecutive iterations. After convergence, the objective function was computed as $L = \Sigma(i=1 \text{ to } k) \Sigma(x \in Di) ||x - \mu i||^2$, where Di is cluster i and μi is its final centroid.

Solution Approach and Results

The same K-means procedure was applied independently for each value of k from 2 through 10. Cluster memberships were recomputed at each iteration, and final assignments were determined using the converged centroids before evaluating the loss. The final objective-function values are shown in Figure 1.

Figure 1. K-Means Strategy 1: Loss vs. Number of Clusters



Observations and Analysis

The objective function decreased for every tested increase in k. The largest reduction occurred from k = 2 to k = 3, where the loss decreased from approximately 2498.11 to 1294.30, a reduction of about 48.2%. The decrease from k = 3 to k = 4 was also substantial at approximately

38.8%. The curve shows a pronounced initial decline and an elbow-like bend around k = 4, after which many of the additional reductions are smaller. The reductions are not uniform. For example, the loss decreases by only about 5.9% from k = 7 to k = 8, while the decrease from k = 9 to k = 10 is larger. This behavior is reasonable because Strategy 1 uses a separate prescribed initialization for every value of k, and K-means can converge to different local solutions depending on its initial centroids. In this particular run, however, the overall loss decreases consistently from k = 2 through k = 10. The total reduction from k = 2 to k = 10 is approximately 92.7%.

Figure 2. K-Means Strategy 1: Final Clustering for k = 10

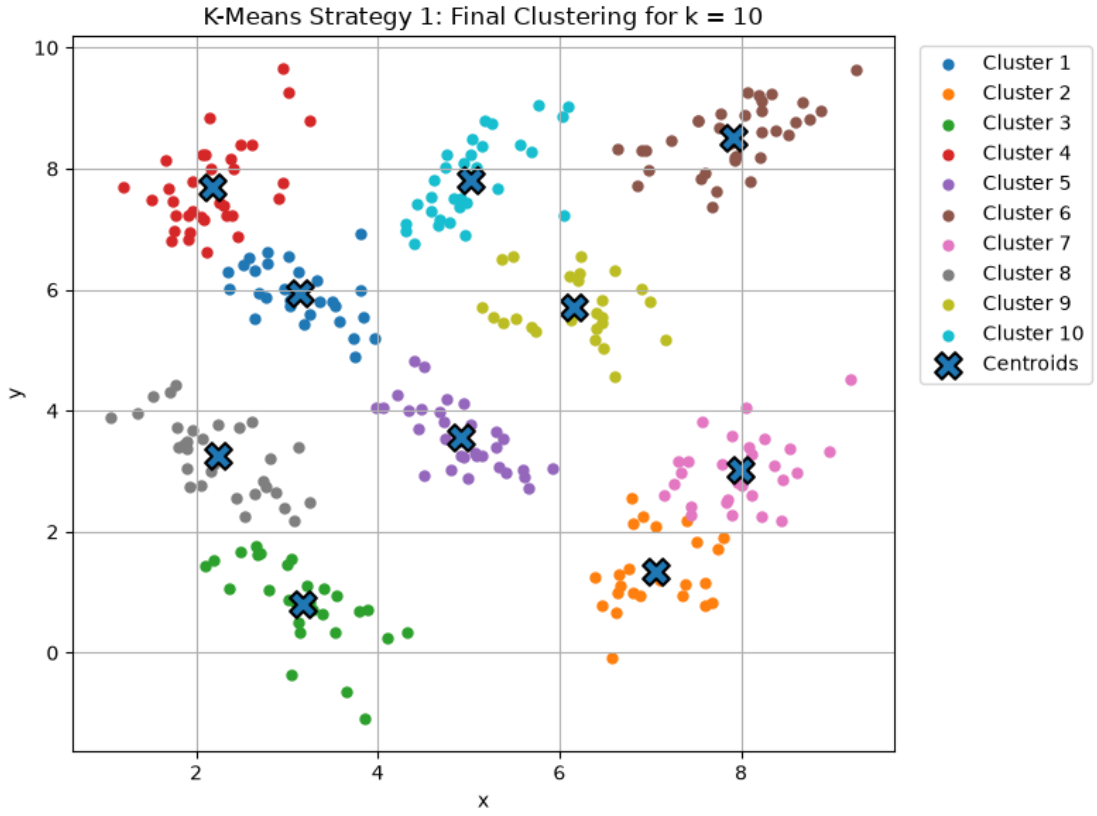


Figure 2 provides a geometric explanation for the decreasing objective function. At low values of k, each centroid must represent a relatively broad region containing multiple visible groups of samples. As k increases, these regions are divided into smaller and more localized clusters. Consequently, the final centroids lie closer to their assigned samples, reducing the total within-cluster squared distance. At k = 10, the clusters closely follow the visible local groupings in the two-dimensional dataset, with each centroid positioned near the center of its assigned cluster. The completed implementation was verified by restarting the kernel and running the entire notebook from the beginning. It was then tested in Ed Lessons, where all 19 autograder test cases passed for a score of 54/54 on the centroid and loss results. The results demonstrate how the number of clusters and the Strategy 1 initialization affect K-means convergence and the resulting within-cluster objective value.